

The concentration problem in single-stock returns

Why most stocks underperform Treasury bills, and what this implies for active management

Abstract

We synthesise the empirical findings of Bessembinder (2018) and subsequent extensions on the cross-sectional distribution of single-stock returns over the long run. Across 25,332 U.S. common stocks listed between 1926 and 2016, only 42.6% generated lifetime returns exceeding the one-month Treasury bill — the median listed stock destroys wealth relative to cash. Of the \$34.8 trillion in lifetime market wealth created by listed U.S. equities, just 86 stocks (0.34% of the population) account for half. The five companies contributing most are Apple, ExxonMobil, Microsoft, GE, and IBM, jointly accounting for 8.3% of total market value creation. The implications for active management are stark: a portfolio that fails to hold the small number of extreme-positive-skewness winners will systematically underperform the index, even if its losers are typical. We extend the analysis through 2021 using updated CRSP data, document the cross-country generalisation by Bessembinder, Chen, Choi, and Wei (2019), and discuss what the findings imply for index investing, concentrated active management, and the equity premium puzzle.

Key findings

1. **Most stocks underperform Treasury bills.** Across 25,332 U.S. listed common stocks 1926–2016, only 42.6% produced lifetime returns above the one-month Treasury bill. The median listed stock destroys wealth relative to cash.
 2. **Half of all wealth creation comes from 0.34% of stocks.** Just 86 stocks of the 25,332 in the sample account for half of all lifetime market wealth created. Apple alone contributes more than 1.5% of the total.
 3. **The cross-country pattern is similar.** Across 64,000 stocks in 42 countries, only 39% beat one-month Treasury bills. The U.S. is at the favourable end of the international distribution.
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4. **Active managers face a structural disadvantage.** If 99.7% of the value-creation is in 1% of the names, an undiversified portfolio without those names underperforms the index in expectation.
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 5. **The case for indexing rests on this finding.** Indexing guarantees ownership of the small number of mega-winners. Concentrated active management requires either correctly identifying winners ex ante or accepting underperformance probability.

EXEC. Executive summary

The empirical equity premium is generated by a tiny minority of stocks. The median stock destroys wealth relative to cash. This finding, established by Bessembinder (2018) and replicated cross-country, is among the most important empirical results in modern asset pricing.

This note synthesises the findings of [Bessembinder \(2018\)](#) and the subsequent literature, with an emphasis on what the results imply for portfolio construction and active management. The headline empirical fact is that, across 25,332 U.S. common stocks listed between July 1926 and December 2016, only 42.6% generated lifetime returns (price plus dividends, compounded over the period the stock was listed) that exceeded the contemporaneous one-month Treasury bill. The remaining 57.4% — a clear majority — destroyed wealth relative to risk-free cash.

The aggregate equity premium is preserved despite this median underperformance because of extreme positive skewness in the right tail. The single best-performing stock in Bessembinder's sample (ExxonMobil at the time of his analysis; Apple in updated extensions) generated lifetime market value creation of approximately \$1 trillion. The top 86 stocks (0.34% of the population) accounted for half of all market value creation. The top 1,092 stocks (4.3%) accounted for all of it; if those names are removed from the universe, the equity premium disappears.

The result has three direct implications for portfolio construction.

First, the case for indexing is stronger than diversification arguments alone suggest. Indexing guarantees that the investor owns the small number of mega-winners that drive aggregate equity returns. A portfolio that excludes them — even by accident, even if its excluded names are statistically representative of the broader market — will underperform the index in expectation.

Second, concentrated active management faces a structural disadvantage. A portfolio of 30 stocks selected from the universe of 25,332 has, by simple combinatorics, a low probability of including the right names. Active managers can overcome this disadvantage by skill in security selection, but the bar is high: the manager must correctly identify the rare extreme winners ex ante.

Third, the equity premium puzzle of [Mehra and Prescott \(1985\)](#) may be reframed. If aggregate returns are driven by a small number of extreme outcomes, the relevant risk premium is compensation for the variance of those tail outcomes — substantially larger than the variance computed from index-level data. The puzzle dissolves partially when single-stock skewness is properly accounted for.

The note proceeds as follows. Section 2 reviews Bessembinder's primary findings. Section 3 covers the cross-country extensions. Section 4 examines updated post-2016 data. Section 5 discusses implications for active management and indexing.

01. The Bessembinder (2018) findings

[Bessembinder's 2018 paper](#) documents the cross-sectional distribution of single-stock long-run returns using CRSP data on the universe of U.S. common stocks listed between July 1926 and December 2016. The paper's empirical contributions are simple to state but counterintuitive in their magnitude.

1.1 Lifetime return distribution

For each stock in the universe, Bessembinder computes the lifetime cumulative return — the total wealth that would have been generated by a \$1 investment held from the stock's first listing date until either delisting or the end of the sample. This is compared to the wealth that would have been generated by a \$1 investment in one-month Treasury bills over the same period.

The headline statistics:

Table 1. Lifetime return statistics, U.S. common stocks 1926–2016

Statistic	Value
Total stocks in sample	25,332
Stocks beating T-bills (lifetime)	10,791 (42.6%)
Stocks underperforming T-bills	14,541 (57.4%)
Median lifetime return	−9% (cumulative, undefined ann.)
Mean lifetime return	+22,000% (cumulative, skew-driven)

The median lifetime return is negative — the median stock loses money over its lifetime in absolute terms. The mean lifetime return is positive and large, but driven entirely by the right tail. The distribution is so heavily right-skewed that the standard mean-versus-median gap of typical financial data is amplified to extraordinary levels.

1.2 Contribution to aggregate market value creation

The aggregate U.S. equity market created approximately \$34.8 trillion in lifetime shareholder value over the 1926–2016 period (Bessembinder's calculation, in 2016 dollars). This is the sum of all positive lifetime contributions; negative contributions are offset by positive ones in the aggregate.

Table 2. Contribution to aggregate market value creation

Stock count	% of sample	Cumulative contribution
Top 1	0.004%	2.6% (ExxonMobil)
Top 5	0.020%	8.3%
Top 50	0.197%	39.3%

Stock count	% of sample	Cumulative contribution
Top 86	0.339%	50.0%
Top 1,092	4.31%	100%

50% of all lifetime market value creation came from 86 of the 25,332 stocks. 100% of the value creation came from 1,092 stocks; the remaining 24,240 stocks (95.7% of the universe) collectively contributed nothing to aggregate wealth creation, with their winners offsetting their losers.

The five top contributors at the time of the original paper were ExxonMobil, Apple, Microsoft, GE, and IBM. Updated through 2021 (see Section 3), the top five become Apple, Microsoft, ExxonMobil, Amazon, and Alphabet — illustrating the ongoing ascendancy of the dominant tech franchises.

1.3 The equity premium decomposition

The aggregate equity premium over Treasury bills is preserved across the full sample at approximately 6% annualised over the 90-year period. Bessembinder's contribution is to show that this aggregate premium is not the typical experience of holding any randomly selected stock. The expected stock — that is, the equal-weighted mean across the universe — produces returns close to T-bills, with the equity premium concentrated in extreme right-tail outcomes.

This is consistent with a positively-skewed stable Paretian distribution rather than a normal distribution. The implication for portfolio construction is that the diversification benefits of multi-stock holding are not just variance reduction (the standard CAPM argument) but the inclusion of the rare extreme winners that drive aggregate returns. A portfolio of 30 stocks chosen at random will, in expectation, include zero of the top-86 contributors. A portfolio of 500 stocks (S&P 500 size) will include perhaps 1–3 of them. A portfolio of 3,000 stocks (Russell 3000 size) will include perhaps 10–20.

02. Cross-country generalisation

[Bessembinder, Chen, Choi, and Wei \(2019\)](#) extend the analysis to a cross-country sample of 64,000 stocks across 42 countries from 1990 to 2018. The qualitative finding generalises: only 39% of stocks beat one-month U.S. Treasury bills over their lifetimes. The U.S. occupies the favourable end of the international distribution; emerging markets in particular show median lifetime returns even more negative than the U.S. average.

Table 3. Cross-country lifetime return statistics, 1990–2018

Country group	Stocks beating T-bills	Top 1% contribution
United States	42%	38%
Developed Markets ex-US	39%	42%
Emerging Markets	36%	47%
Global Aggregate	39%	42%

The pattern of extreme right-skewness — most stocks underperforming, a tiny minority of stocks accounting for the majority of value creation — is universal across the 42 countries studied. The specific magnitude varies: emerging markets show even greater concentration, with the top 1% of stocks contributing 47% of total wealth creation versus 38% in the U.S.

The implication is that the indexing case is stronger in concentrated markets, not weaker. In emerging markets where the right-tail concentration is highest, the cost of failing to own the few extreme winners is greatest. The intuition that emerging markets favour active stock selection is not supported by this distributional fact.

03. Updated post-2016 results

The 2018 paper's sample ends in December 2016. Several developments since then have shifted the top-contributor rankings substantially. We update key statistics through December 2021 using CRSP data extensions and supplementary tabulations ([Bessembinder, 2023](#)).

Table 4. Top 10 lifetime contributors, U.S. common stocks 1926–2021

Rank	Company	Lifetime value created (\$B)
1	Apple	\$2,670
2	Microsoft	\$1,910
3	Amazon	\$1,330
4	Alphabet (Google)	\$1,250

Rank	Company	Lifetime value created (\$B)
5	ExxonMobil	\$1,120
6	Berkshire Hathaway	\$890
7	Johnson & Johnson	\$810
8	Walmart	\$770
9	Procter & Gamble	\$700
10	JPMorgan Chase	\$650

The 2016–2021 period saw extraordinary value creation by the FAANG/MAANG group, with Apple's contribution alone roughly tripling over five years to approximately \$2.7 trillion. The displacement of ExxonMobil from #1 to #5 and the rise of four tech firms into the top ranks reflects the dominant economic story of the 2010s.

Notably, the top-86 threshold (50% of cumulative value creation) has shifted only modestly over the post-2016 period. Approximately 80–90 stocks continue to account for half of all lifetime wealth creation — the concentration property is structural, not period-specific. The identities of the names rotate; the concentration does not.

3.1 The 2020–2021 retail trading episode

The 2020–2021 retail trading boom — characterised by zero-commission trading, social-media-driven concentrated positions in single stocks, and substantial capital inflows from individual investors — provides a natural experiment for testing the implications of Bessembinder's findings. Several "meme stocks" (GameStop, AMC) generated short-period returns that, viewed in isolation, suggested concentrated single-stock investing could produce extraordinary outcomes.

The longer-horizon implications are more sobering. Of the approximately 200 stocks that experienced retail-driven outperformance during the 2020–2021 period, the median 12-month subsequent return was approximately –45%. The right-skew that drives long-run wealth creation in indexed equities is not present at the level of individual retail-favourite names, which exhibit short-term skew without long-term skew. The Bessembinder framework correctly predicts that retail concentration in 2020–2021 names will, on average, underperform passive index investment.

04. Implications

4.1 The strengthened case for indexing

The classic case for indexing rests on three arguments: (a) costs are lower, (b) skill is rare among active managers, (c) diversification reduces variance. Bessembinder's findings add a fourth, mechanically more compelling argument: indexing guarantees inclusion of the rare extreme winners that account for aggregate equity returns.

The fourth argument is conceptually different from the first three. The cost and skill arguments are statistical: in the aggregate, after costs, active managers tend not to outperform. The diversification argument is variance-related: holding more stocks reduces the noise around the expected return. The Bessembinder argument is structural: the expected return itself is concentrated in a small number of names, and any portfolio that fails to include them will systematically underperform regardless of cost or skill considerations.

The implication is that indexing is not just one reasonable approach to equity investing among many; it is the approach that mechanically captures the entire equity premium. Active managers who underweight or omit the dominant value creators are taking on negative-expected-return tracking error that requires extraordinary skill to overcome.

4.2 The bar for active management

Active management can outperform index investing if and only if the manager systematically identifies the extreme winners ex ante. This is a higher bar than is typically appreciated. A manager who beats the index by selecting "good" stocks of average performance characteristics will not produce sustained outperformance, because the index's returns are not driven by average stocks.

The skills required for sustained outperformance are therefore not the skills traditionally associated with security analysis (analysing financial statements, evaluating management quality, modelling cash flows). They are the skills associated with identifying extreme positive optionality — companies with unusually large potential upside, even at the cost of some downside risk. [Mauboussin \(2017\)](#) documents that successful long-run active managers tend to share this orientation: they are willing to hold concentrated positions in extreme-skew names even at the cost of higher tracking error.

4.3 Equity premium reframing

The [Mehra-Prescott \(1985\)](#) equity premium puzzle asked why investors demand a 6% premium for holding equities given the relatively modest variance of aggregate returns. The puzzle was framed in terms of the standard mean-variance Sharpe ratio: the realised premium implies a level of risk aversion that seemed implausibly high.

Bessembinder's findings reframe the puzzle. If aggregate equity returns are generated by extreme right-tail outcomes in single stocks, the relevant risk metric is not the variance of aggregate returns but the

cross-sectional skewness of the wealth creation distribution. The equity premium is compensation for the willingness to hold an asset class in which most individual realisations destroy wealth, with success contingent on capturing the small number of extreme positive outcomes. Under this framing, the premium magnitude is more plausible.

4.4 Concentrated portfolios

Some sophisticated investors — Berkshire Hathaway being the canonical example — hold concentrated portfolios of 10–30 names and outperform indexes over long horizons. This is not inconsistent with Bessembinder's findings; it is consistent with the small number of investors who possess the skill to identify extreme winners ex ante. The implication is not that concentrated portfolios cannot work — they can, with sufficient skill — but that the burden of proof on a concentrated portfolio is to demonstrate skill above the high bar set by the empirical concentration property.

05. Discussion and conclusion

The Bessembinder findings are among the most important empirical contributions to asset pricing in the past two decades. They alter the case for indexing in a fundamental way: from a statistical preference (costs, skill, diversification) to a structural mechanical fact (you must own the names that drive the aggregate return). They alter the bar for active management from "outperform some benchmark" to "correctly identify the extreme winners." They reframe the equity premium puzzle in a way that partially dissolves it.

Limitations. The findings are based on listed-company data and exclude private equity, where the wealth creation distribution may have different properties. The lifetime-return measure is sensitive to listing duration; firms that are listed briefly contribute less even if their per-period returns are strong. The cross-country extensions cover 1990–2018 only; pre-1990 international data is sparse. Most importantly, Bessembinder's data is U.S.-listed, which excludes some major global firms that matter for global wealth creation but happen to be listed in their home jurisdictions.

Conclusion. The median U.S. listed common stock destroys wealth relative to Treasury bills. The aggregate equity premium is generated by approximately 0.34% of stocks. The implication for portfolio construction is that indexing — or active strategies that explicitly target extreme positive optionality — captures the equity premium reliably; concentrated active strategies face a structural disadvantage that requires unusual skill to overcome. The findings dissolve, partially, the equity premium puzzle of [Mehra and Prescott \(1985\)](#) by reframing the relevant risk as cross-sectional skewness rather than aggregate variance. They are among the most consequential empirical findings of the past two decades for individual and institutional asset allocation.

A randomly selected stock loses to T-bills. The aggregate equity premium is generated by 0.34% of stocks. Indexing is not a statistical preference — it is the

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stock returns

| *only reliable way to own that 0.34%.*

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APX. Appendix A: Sensitivity to methodology

A.1 Lifetime versus holding-period returns

Bessembinder's headline statistics use lifetime returns: the cumulative return from first listing to delisting or sample end. An alternative methodology computes monthly returns and equal-weights them across stocks. The two methodologies produce different qualitative pictures.

Table A1. Lifetime returns vs equal-weighted monthly returns

Methodology	Outperformance vs T-bills
Lifetime cumulative	42.6% of stocks
Equal-weighted monthly	~52% of stock-months
Capitalisation-weighted monthly	~58% of stock-months

Lifetime cumulative returns produce the lowest "winner rate" because the compounding interaction with negative returns is asymmetric: a 50% loss followed by a 50% gain leaves the investor at 75% of starting value, not 100%. Equal-weighted monthly returns are closer to a fair coin flip. Bessembinder's choice of lifetime methodology is appropriate for evaluating the actual experience of holding a stock through its lifetime, which is the experience an investor has.

A.2 Distinction from positive expected returns

Bessembinder's findings do not contradict the existence of a positive expected return on equities. The expected return — the population mean — is positive and substantial. The findings show only that the population mean is generated by extreme outcomes in a small minority of names, not by typical performance across the full population. The distinction between "expected return is positive" and "most realisations are positive" is the heart of the contribution.

A.3 Index inclusion bias

The S&P 500, in particular, has a structural advantage in capturing extreme winners: stocks that grow large enough to enter the S&P 500 are by definition likely to be among the better historical performers. Index inclusion criteria therefore amplify the concentration property in published index returns. Total-market indexes (Russell 3000, MSCI USA Total) have somewhat less extreme concentration than the S&P 500 because they include smaller, more variable names.

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PUBLISHED BY

ProRealAlgos Research
prorealalgos.com

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REVISION HISTORY

v1.0	JAN 2022	Initial publication.
v1.1	SEP 2023	Updated Bessembinder concentration figures with newer working-paper data. Magnitudes shifted by less than 2 percentage points.

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